SUMMARY · MK 2 FINANCIAL MANAGEMENT & STRATEGY · WEEK 2

# Financial Statement & Ratio Analysis

Week 1 taught reading statements; Week 2 teaches analyzing them. Reporting supplies information; analysis turns it into judgment. You learn a 6-step framework, common-size analysis, five ratio families and DuPont.

## Material

### 1 Reporting vs analysis

**Reporting** supplies information; **analysis** assesses past and likely future performance for capital-allocation decisions. Analysts ask whether results beat expectations, what drove them, whether margins last and what could reverse them.

Read the **notes** (where real analysis often begins), management commentary (critically) and the audit report (fair presentation, not a guarantee of success). Use top-down logic: macro → industry → company.

### 2 Earnings quality: profit isn't always performance

Reported profit can diverge from economics via omitted value, method choices, estimates and earnings management. Ask: sustainable, cash-backed, recurring, consistent with fundamentals?

### 3 6-step framework and tools

1. Purpose
2. Data
3. Process

4. Analyze
5. Conclude
6. Follow up

**Computation isn't analysis** — a ratio is an indicator. Tools: graphics, regression, common-size (vertical: item ÷ base; horizontal: current ÷ base – 1), ratios.

**Red flag:** revenue +19%, receivables +38%, inventory +58% → loose credit, slow collection, aggressive recognition, obsolete stock or channel stuffing. Communicate starting from the decision question.

## 4 Five ratio families

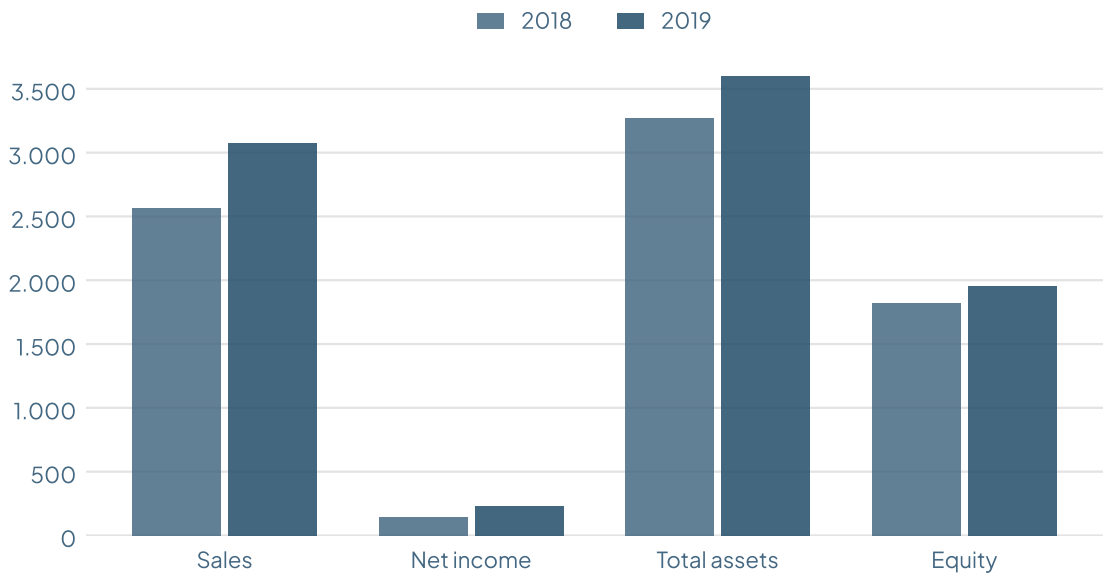
Users: shareholders, creditors, management. Compare via trend, cross-section, benchmarking; beware industry, size, seasonality, policies, one-offs.

**Liquidity:** current ratio, quick ratio — too high may mean idle capital. **Activity:** inventory turnover, collection period, asset turnover. **Solvency:** debt ratio, D/E, interest coverage — leverage lifts ROE and risk. **Profitability:** gross/operating/net margin, ROA, ROE. **Valuation:** P/E, P/CF, P/S, P/BV. The families are linked.

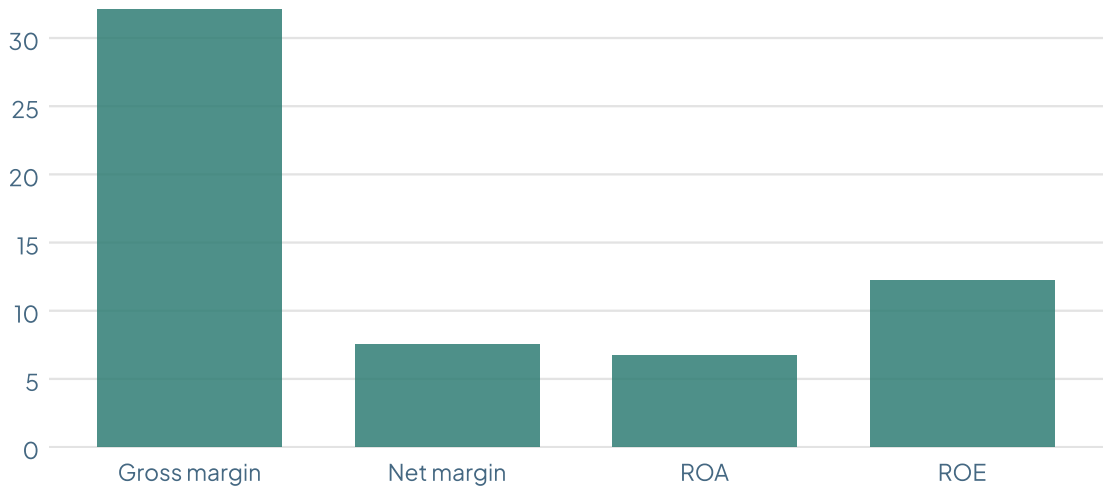
## 5 Bartlett's ratios charted

Deep dive

### Bartlett: 2018 vs 2019 ($000)




### Bartlett profitability 2019 (%)




**1.97x**
Current ratio

**1.51x**
Quick ratio

**45.7%**
Debt ratio

**7.09x**
Inventory turnover (~51 days)

**Quick read**

Sales +20%, net income +56% → margin up from 5.8% to 7.5%.

## 6 DuPont analysis: decomposing ROE

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ROE = Net margin × Asset turnover × Financial leverage
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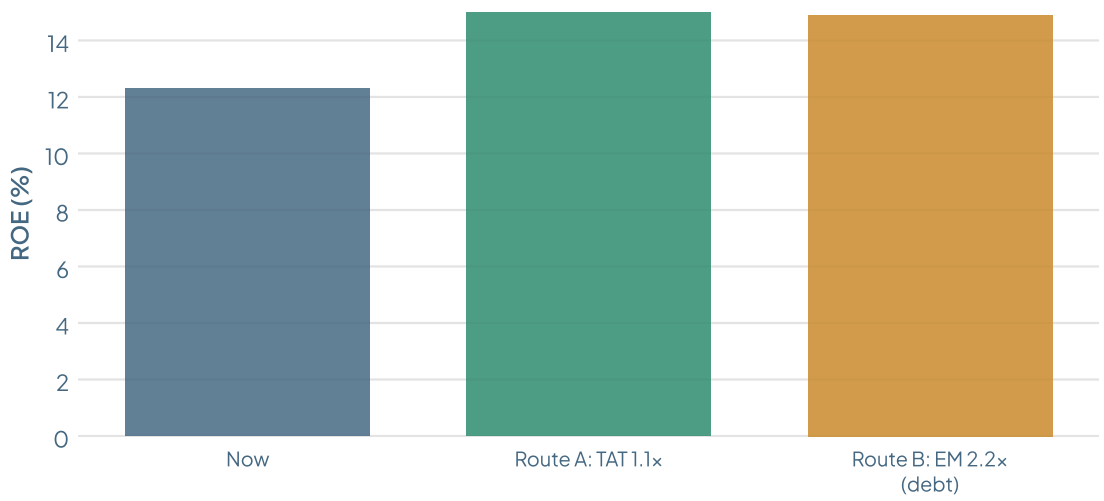
| | A | B | C |
|---|---|---|---|
| ROE | 20% | 20% | 20% |
| Margin | 10% | 5% | 3% |
| Turnover | 2.0× | 2.0× | 4.45× |
| Leverage | 1.0× | 2.0× | 1.5× |

A wins on margin, C on efficiency, B relies on debt — high ROE driven by leverage is fragile. Signals → actions: falling margin (pricing, COGS), falling turnover (capacity, inventory, collections), rising leverage (cash flow, covenants), rising ROE (decompose before celebrating). IFSA's 5-way DuPont splits margin into tax burden, interest burden and EBIT margin.

## 7 Try it: the DuPont calculator Deep dive

Defaults are Bartlett's (NPM 7.5%, TAT 0.90×, EM 1.82× → ROE ≈ 12.2%). Which route to 15% ROE is healthier?

**Two routes to ~15% ROE**




A: 7.5% × 1.1 × 1.82; B: 7.5% × 0.9 × 2.2. Same result, different risk.

**! High ROE isn't always good**

Debt-driven ROE is fragile.

## 8 Ratios in forecasting (bridge to Week 3)

Historical ratios drive forecast models (sales growth, margins, receivable/inventory/payable days, leverage). Don't extrapolate mechanically — combine with strategy, industry and macro assumptions.

## Key takeaways

1. Reporting = information; analysis = judgment for capital allocation.
2. The notes are often where analysis really begins.
3. Computing isn't analyzing — ratios need context.
4. Vertical/horizontal common-size; watch receivables/inventory outgrowing sales.
5. Five ratio families: liquidity, activity, solvency, profitability, valuation.
6. DuPont: ROE = margin × turnover × leverage; leverage-driven ROE is fragile.
7. Forecasting uses historical ratios plus strategy, industry and macro judgment.

## Case studies

Case & worked answers

### Bartlett Company: one-page diagnosis

**Data ($000):** sales 3,074 (2,567); GP 986; NI 231 (148); CA 1,223; CL 620; TA 3,597 (3,270); TL 1,643; equity 1,954 (1,820).

CR 1.97×, QR 1.51×, inventory turnover 7.09×, debt ratio 45.7%, gross margin 32.1%, net margin 7.5%, ROA 6.7%, ROE 12.2%. DuPont: 7.5% × 0.90 × 1.82 ≈ 12.2%.

**Diagnosis:** sales +~20%, NI +~56% → better margins; healthy liquidity; moderate leverage. ROE is margin-driven, not debt-driven — good quality; asset turnover (0.9×) is the improvement area. Ask management what drove margins and whether new assets are fully productive; benchmark against industry and check operating cash flow.

**Discussion:** lenders read cash flow and balance sheet first; equity investors income and cash flow; profitable firms can be illiquid when profit is tied up in

receivables/inventory; ROE can rise with risk via leverage; EPS/P-E and inventory turnover need the notes.

## Worked examples

**Problem 1** Easy **Likuiditas**

Aset lancar Rp600 juta terdiri dari kas 80, surat berharga 40, piutang 180, dan persediaan 300. Liabilitas lancar Rp400 juta. Hitung current ratio dan quick ratio, lalu jelaskan artinya.

1. `Current ratio = Aset lancar ÷ Liabilitas lancar = 600 ÷ 400 = 1,5`
2. `Quick ratio = (Kas + Surat berharga + Piutang) ÷ Liabilitas lancar = (80 + 40 + 180) ÷ 400 = 0,75`
3. Bandingkan: current ratio tampak sehat (1,5×), tetapi separuh aset lancar adalah persediaan. Tanpa persediaan, perusahaan hanya punya Rp0,75 untuk setiap Rp1 utang jangka pendek.

**Answer:** Current ratio 1,5×; quick ratio 0,75×. Likuiditas bergantung pada kemampuan menjual persediaan.

Problem 2 Medium **DuPont**

Penjualan Rp2.000 juta, laba bersih Rp150 juta, total aset Rp1.600 juta, ekuitas Rp800 juta. Pecah ROE dengan DuPont.

1. Net profit margin = 150 ÷ 2.000 = **7,5%**.
2. Perputaran aset (TAT) = 2.000 ÷ 1.600 = **1,25×**.
3. Equity multiplier (leverage) = 1.600 ÷ 800 = **2,0×**.
4. 
```
ROE = 7,5% × 1,25 × 2,0 = 18,75%
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Cek langsung: 150 ÷ 800 = 18,75% ✓
5. ROA = 7,5% × 1,25 = 9,375%. Leverage menggandakan ROA menjadi ROE.

**Answer:** ROE 18,75% = margin 7,5% × perputaran 1,25 × leverage 2,0.

Problem 3 Easy **Solvabilitas**

EBIT Rp300 juta, beban bunga Rp60 juta, total liabilitas Rp800 juta, total aset Rp1.600 juta, ekuitas Rp800 juta. Hitung interest coverage, debt ratio, dan debt-to-equity.

1. Interest coverage = EBIT ÷ bunga = 300 ÷ 60 = **5×** (laba operasi 5 kali lipat bunga).
2. Debt ratio = 800 ÷ 1.600 = **50%**.
3. Debt-to-equity = 800 ÷ 800 = **1,0×**.

**Answer:** Coverage 5×, debt ratio 50%, D/E 1,0×.

## Practice questions

Try answering first. The answer key is below each question.

**1. NPM 7.5%, TAT 0.9×, EM 1.82×. ROE ≈ …**

A. 6.8%
B. 12.3%
C. 15.0%
D. 8.2%

**Answer: B.** 6.75% × 1.82 ≈ 12.3%.

**2. The quick ratio differs from the current ratio because it…**

A. Adds fixed assets
B. Excludes inventory
C. Adds long-term debt
D. Uses net income

**Answer: B.** Inventory is least liquid.

**3. A profitable firm is short of cash. Most likely cause…**

A. Margins too high
B. Accrual profit tied up in receivables and inventory
C. Too little debt
D. Zero tax

**Answer: B.** Profit ≠ cash.

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**4. Creditors look first at…**

A. EPS growth
B. Liquidity, leverage, cash flow
C. Share price
D. Headcount

**Answer: B.** They want to be repaid.

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**5. ROE rose because the equity multiplier jumped. This signals…**

A. Better operations
B. Higher financial risk from debt
C. Better margins
D. Faster inventory

**Answer: B.** Higher EM = more debt.

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**Sources**

- -P3MD- MK 02 Week 2.pdf
- Bab yang dirangkum di deck: IFSA ch.1 & ch.7 (Robinson dkk., Financial Analysis 3rd ed.), Zutter & Smart ch.3